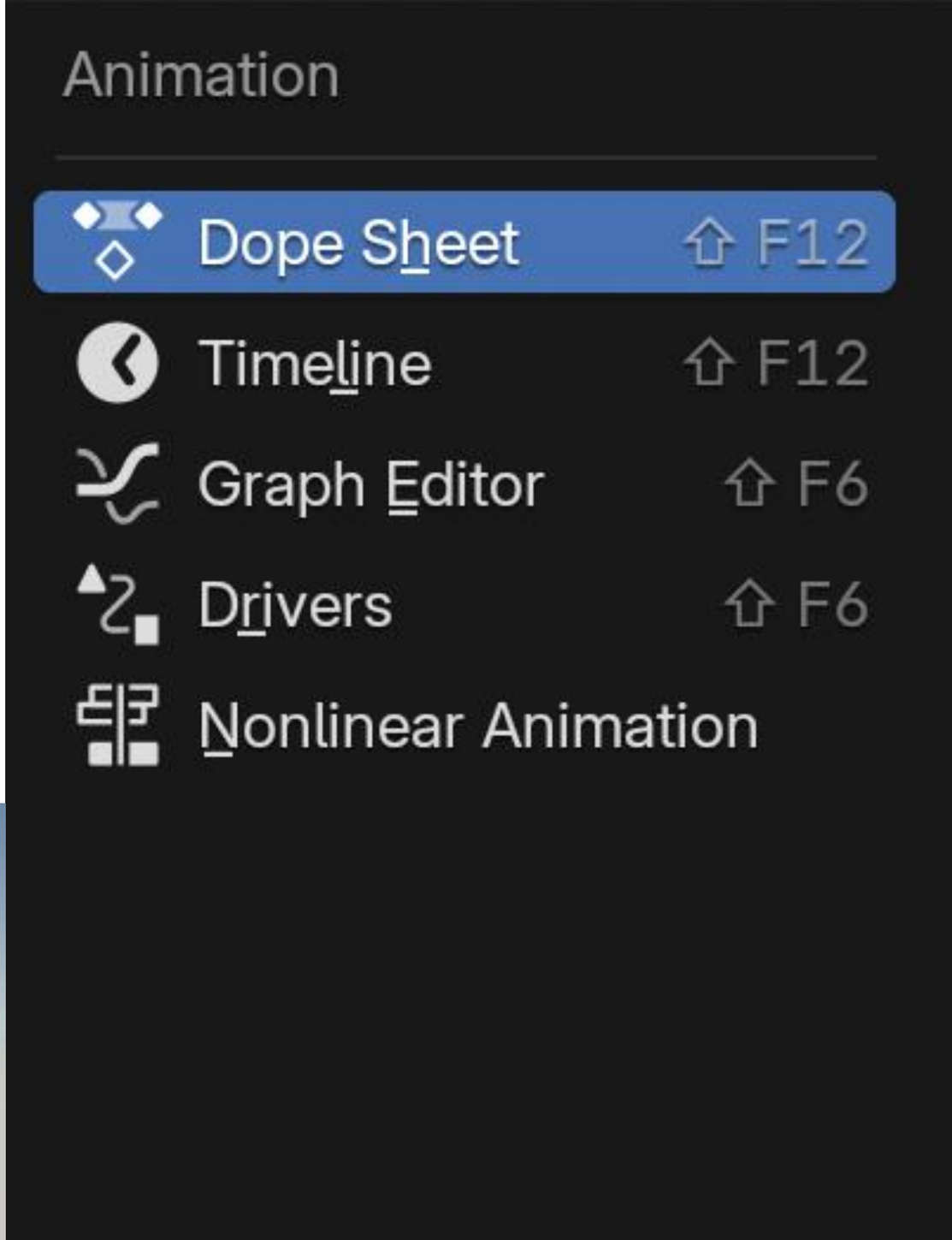
Animation Timeline

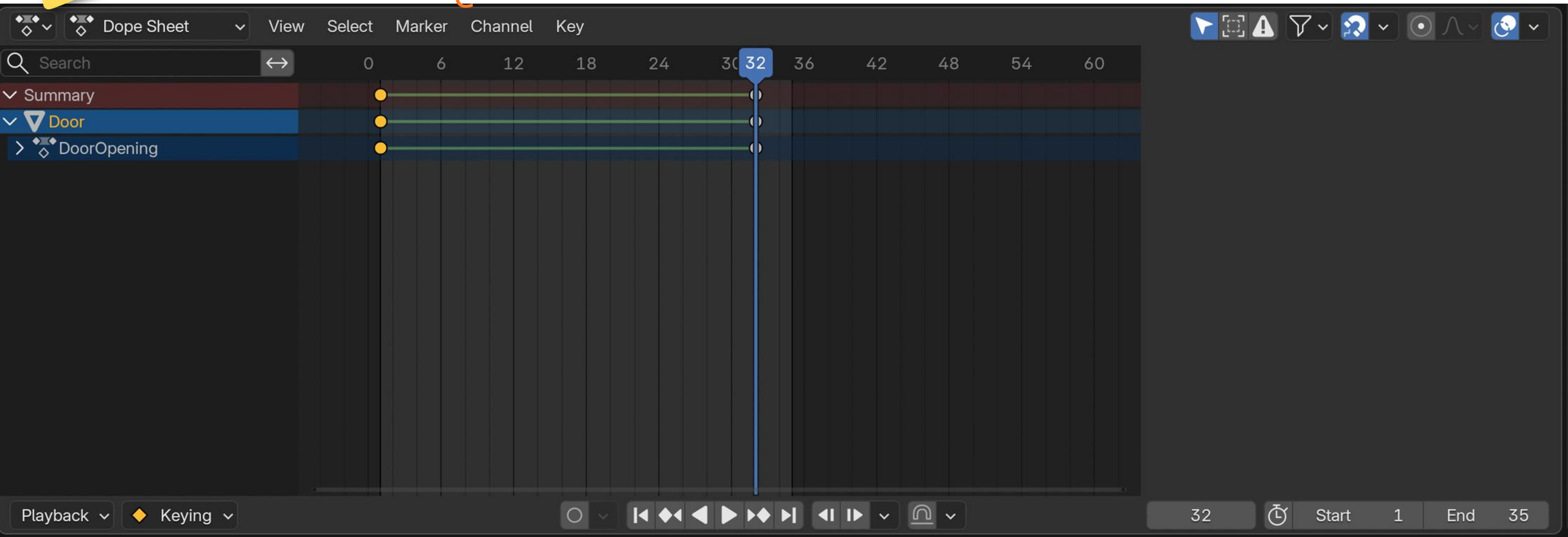
Keyframe & Interpolation



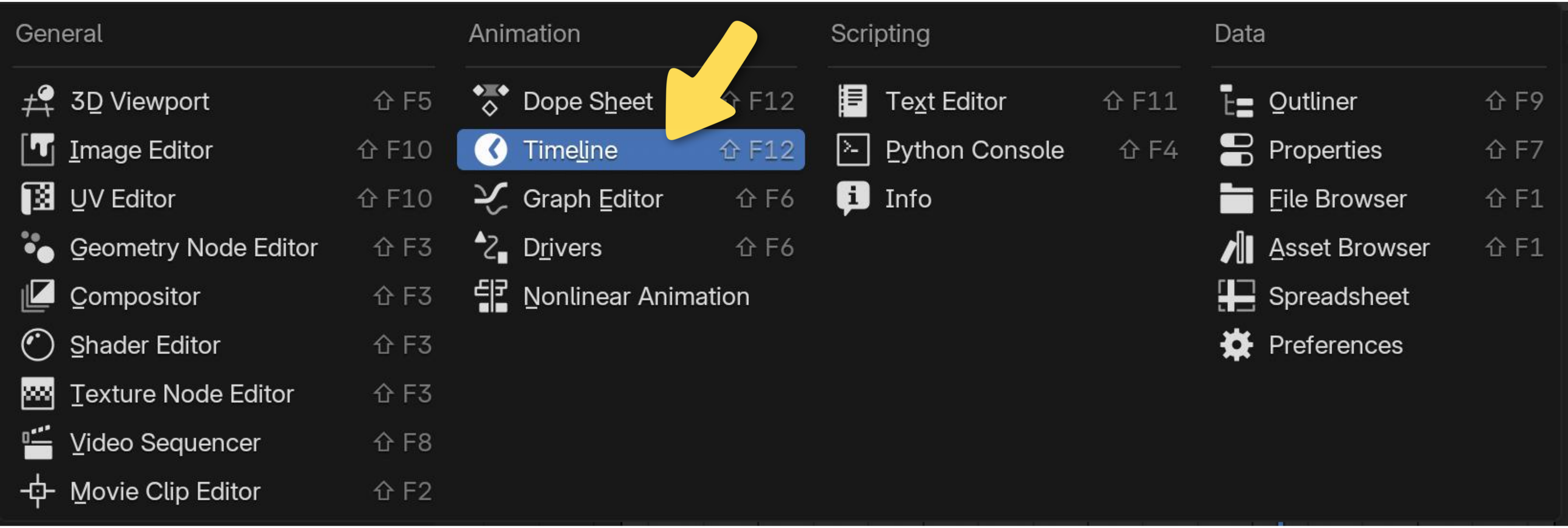


Animation Timeline

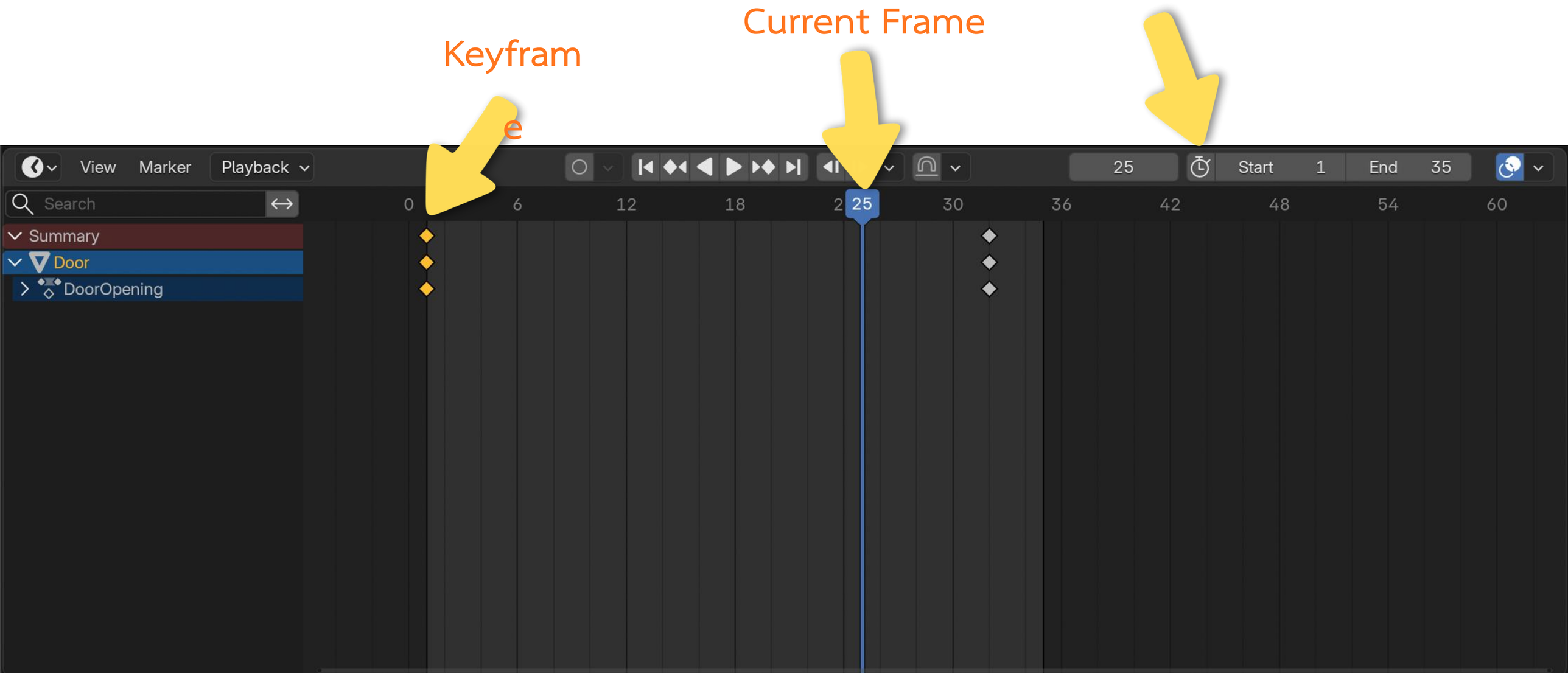
Keyframe



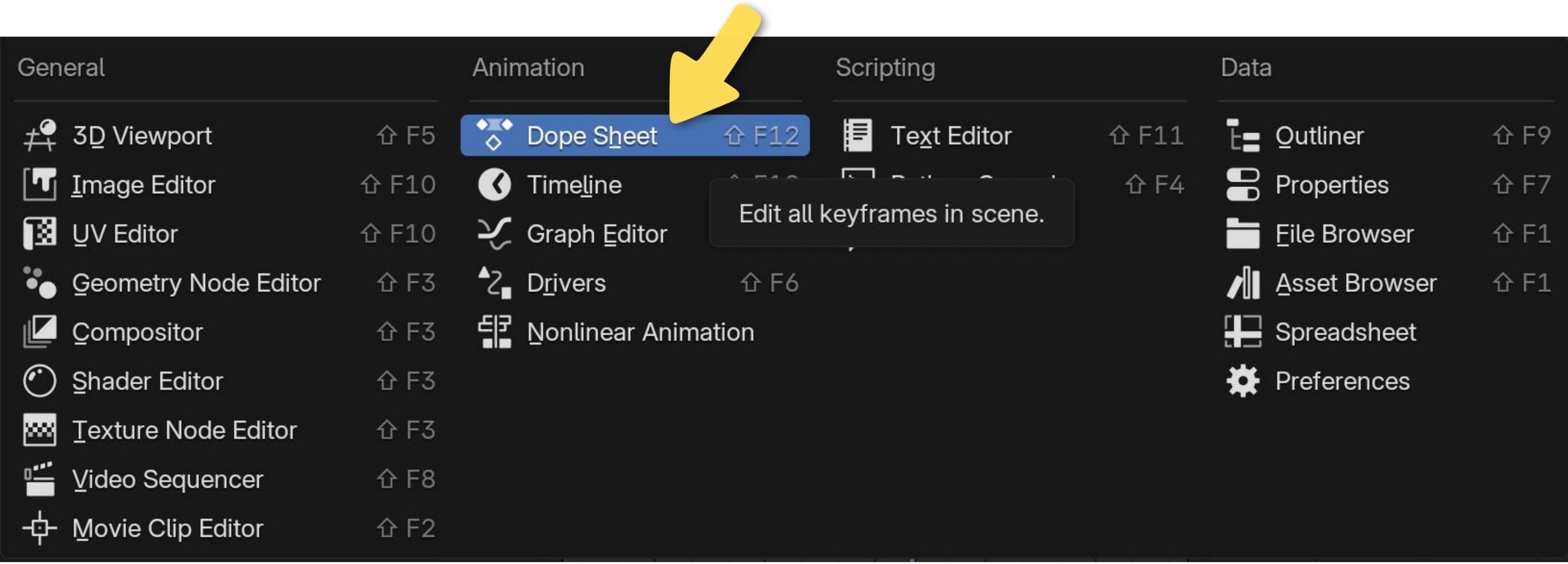
Timeline


General
3D Viewport ⇧ F5
Image Editor ⇧ F10
UV Editor ⇧ F10
Geometry Node Editor ⇧ F3
Compositor ⇧ F3
Shader Editor ⇧ F3
Texture Node Editor ⇧ F3
Video Sequencer ⇧ F8
Movie Clip Editor ⇧ F2
Animation
Dope Sheet ⇧ F12
Timeline ⇧ F12
Graph Editor ⇧ F6
Drivers ⇧ F6
Nonlinear Animation
Scripting
Text Editor ⇧ F11
Python Console ⇧ F4
Info
Data
Outliner ⇧ F9
Properties ⇧ F7
File Browser ⇧ F1
Asset Browser ⇧ F1
Spreadsheet
Preferences

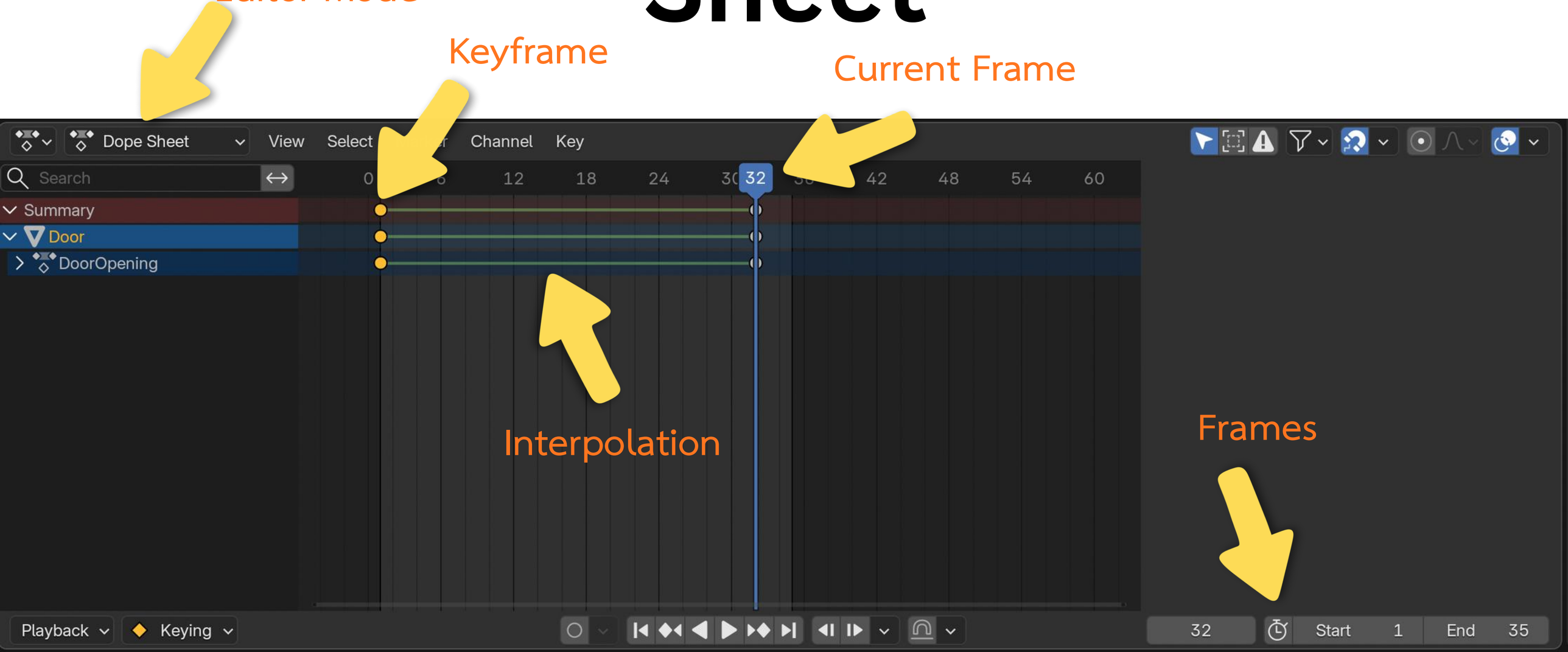
Dope Sheet


Keyfram
e
Current Frame
View
Marker
Playback
25
Start
1
End
35
Search
0
6
12
18
25
30
36
42
48
54
60
Summary
Door
DoorOpening

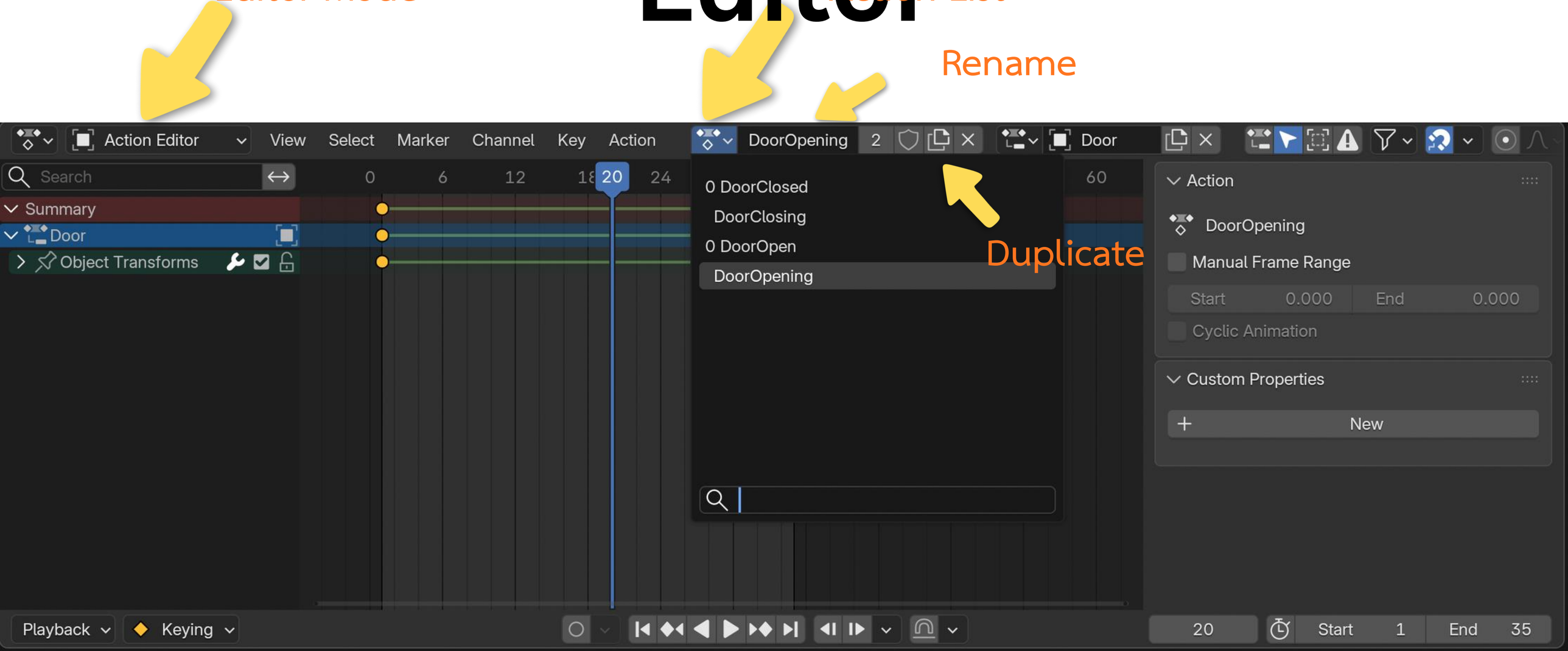
Dope Sheet


General
3D Viewport ⇧ F5
Image Editor ⇧ F10
UV Editor ⇧ F10
Geometry Node Editor ⇧ F3
Compositor ⇧ F3
Shader Editor ⇧ F3
Texture Node Editor ⇧ F3
Video Sequencer ⇧ F8
Movie Clip Editor ⇧ F2
Animation
Dope Sheet ⇧ F12
Timeline
Graph Editor
Drivers ⇧ F6
Nonlinear Animation
Edit all keyframes in scene.
Scripting
Text Editor ⇧ F11
⇧ F4
Data
Outliner ⇧ F9
Properties ⇧ F7
File Browser ⇧ F1
Asset Browser ⇧ F1
Spreadsheet
Preferences

Dope Sheet

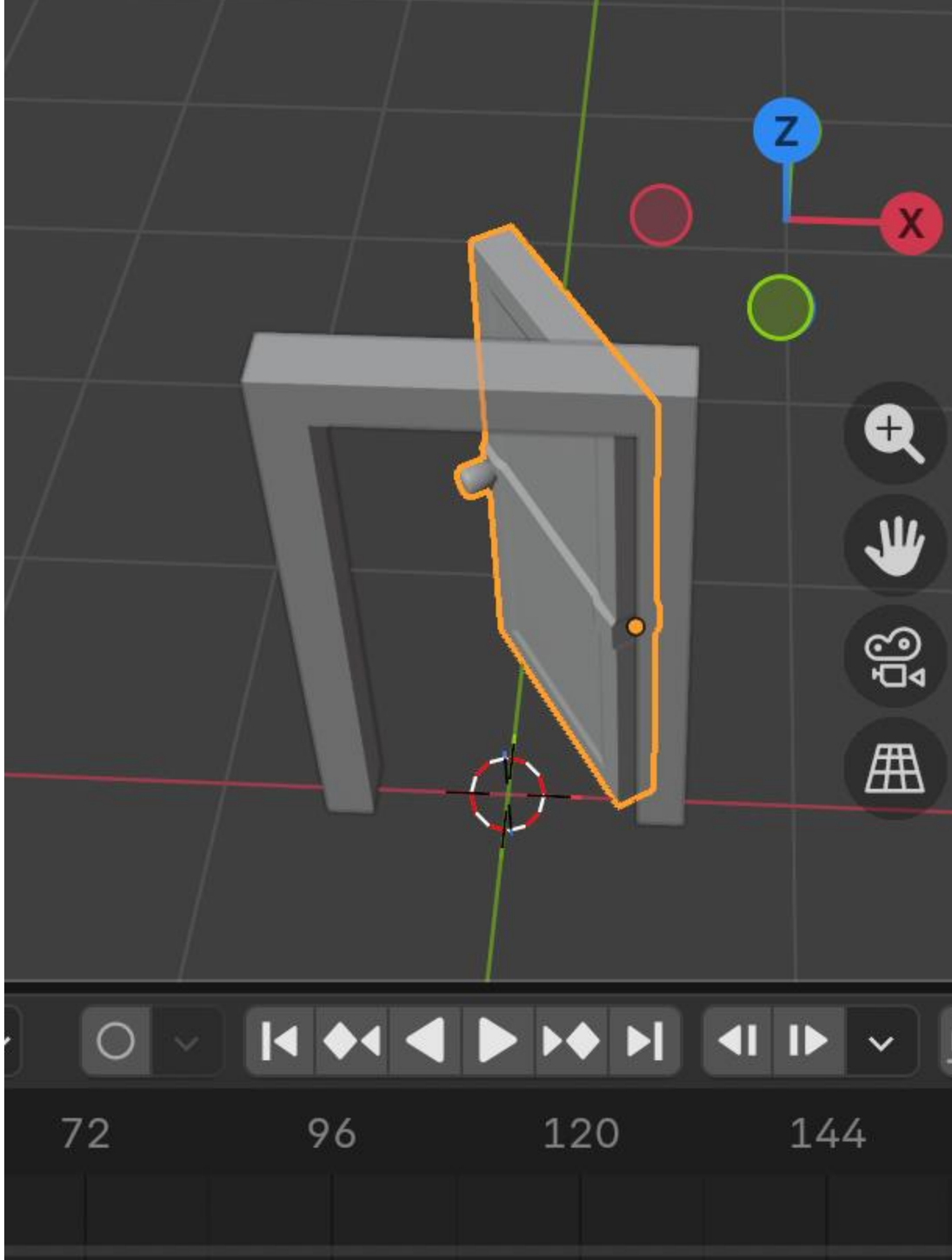

Editor Mode
Keyframe
Current Frame
Dope Sheet
View
Select
Channel
Key
Search
0
12
18
24
30
32
42
48
54
60
Summary
Door
DoorOpening
Interpolation
Frames
Playback
Keying
32
Start
1
End
35

Action Editor

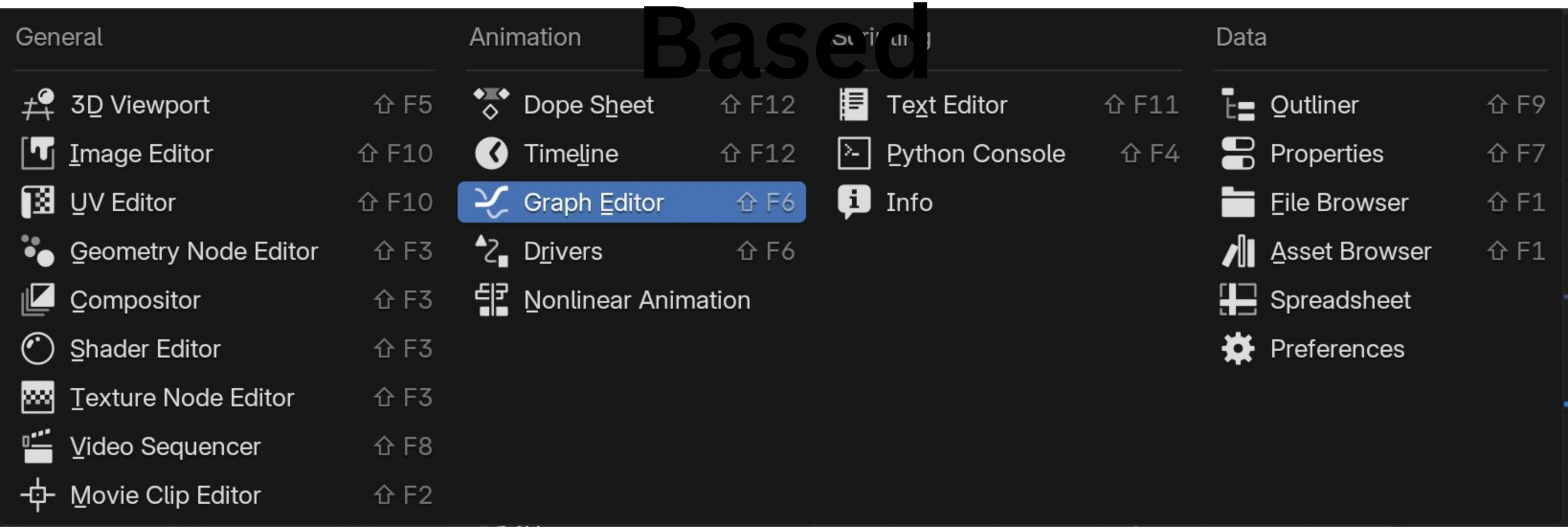


Simple Keyframe Animation

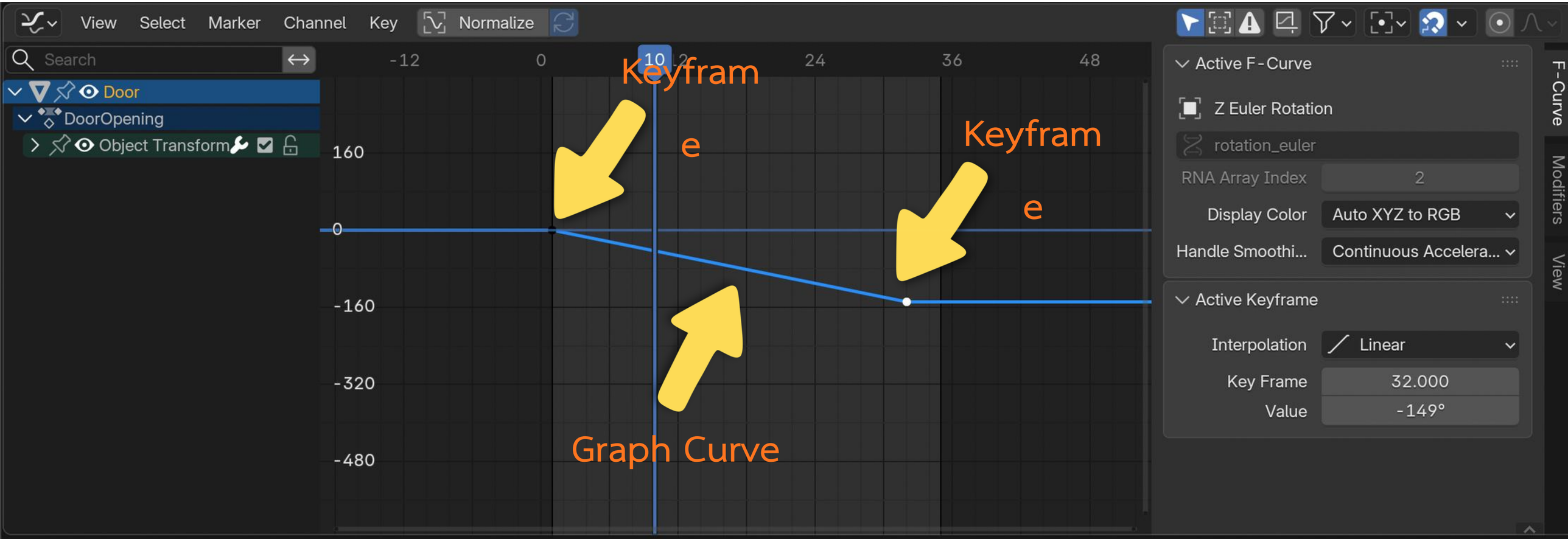
- Model a simple door & animate:
  - Door | Open
  - Door | Closed
  - Door | Opening
  - Door | Closing
- Export to Game Engine



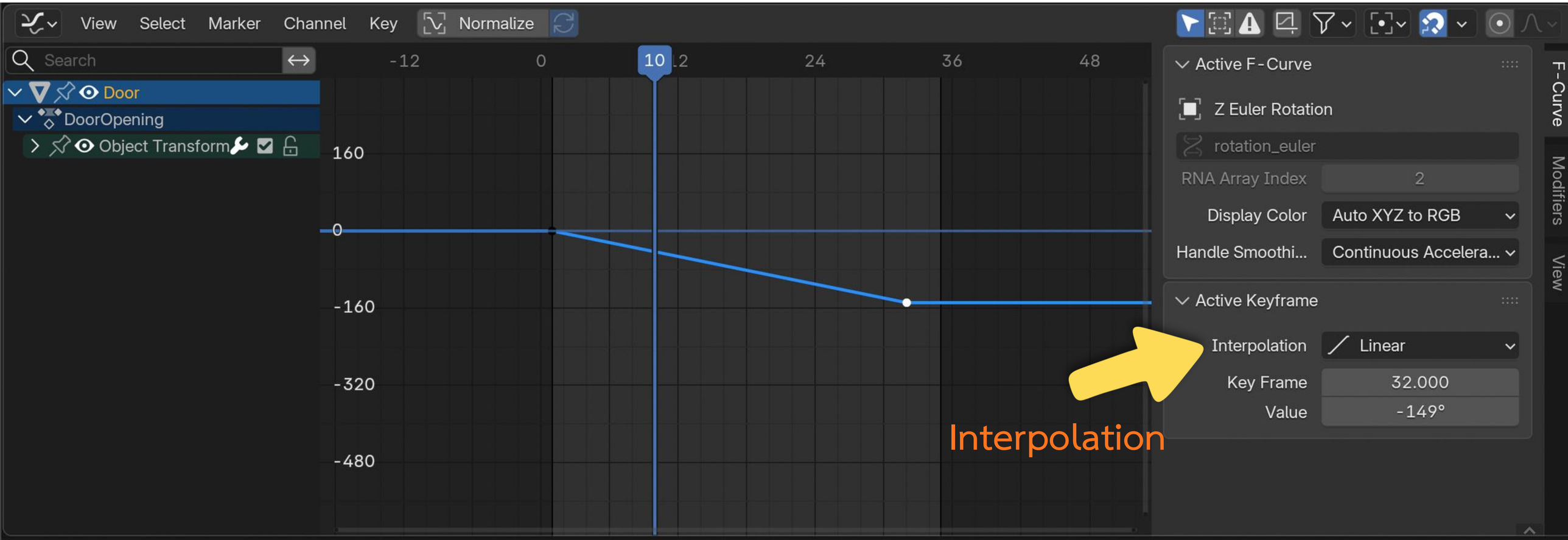
Graph Based


General
3D Viewport ⇧ F5
Image Editor ⇧ F10
UV Editor ⇧ F10
Geometry Node Editor ⇧ F3
Compositor ⇧ F3
Shader Editor ⇧ F3
Texture Node Editor ⇧ F3
Video Sequencer ⇧ F8
Movie Clip Editor ⇧ F2
Animation
Dope Sheet ⇧ F12
Timeline ⇧ F12
Graph Editor ⇧ F6
Drivers ⇧ F6
Nonlinear Animation
Text Editor ⇧ F11
Python Console ⇧ F4
Info
Data
Outliner ⇧ F9
Properties ⇧ F7
File Browser ⇧ F1
Asset Browser ⇧ F1
Spreadsheet
Preferences

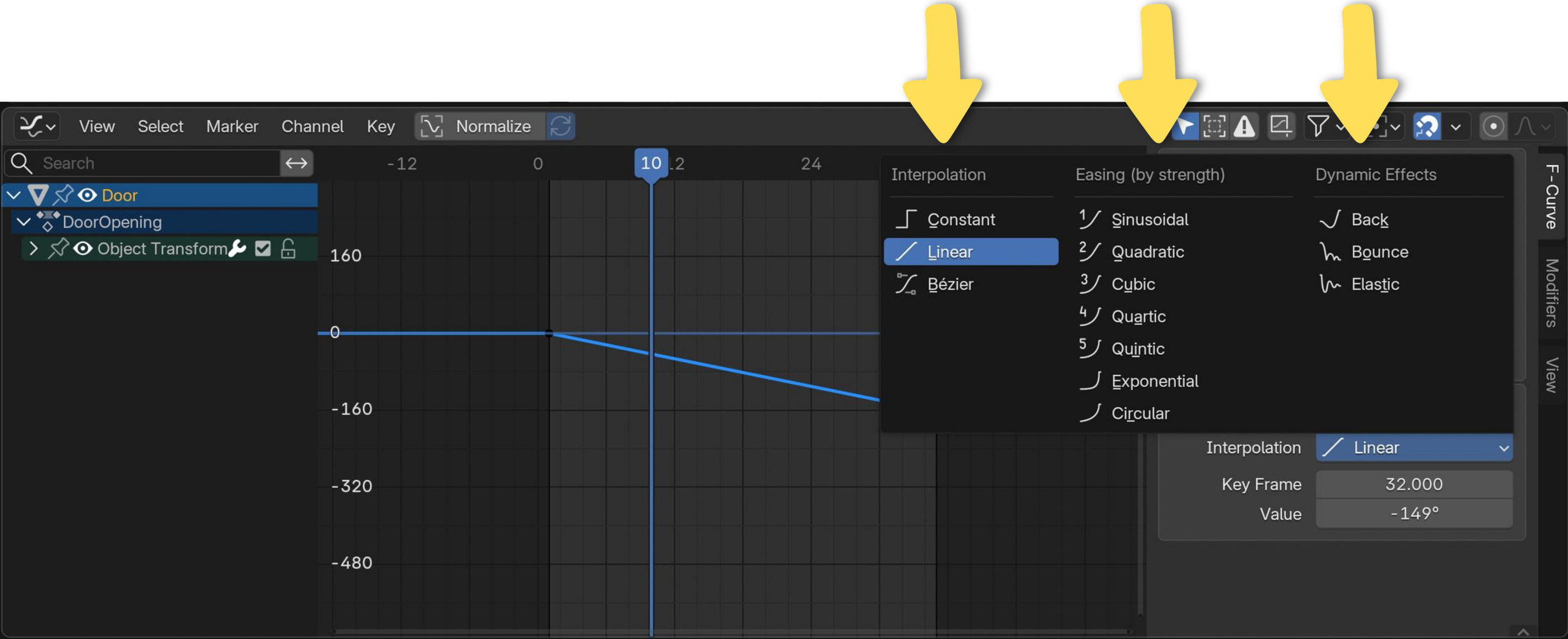
F-Curve


View
Select
Marker
Channel
Key
Normalize
Search
Door
DoorOpening
Object Transform
-12
0
10
24
36
48
160
0
-160
-320
-480
Keyfram
e
Keyfram
e
Graph Curve
Active F-Curve
Z Euler Rotation
rotation_euler
RNA Array Index
2
Display Color
Auto XYZ to RGB
Handle Smoothi...
Continuous Accelera...
Active Keyframe
Interpolation
Linear
Key Frame
32.000
Value
-149°
F-Curve
Modifiers
View

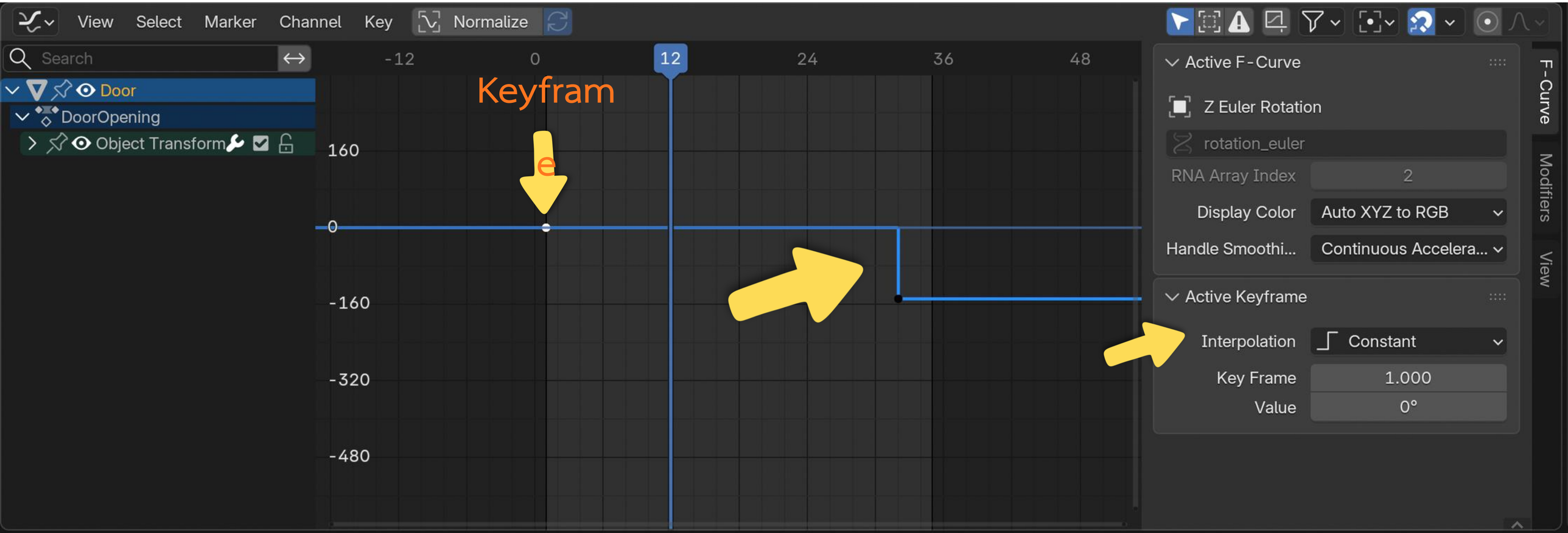
Interpolation



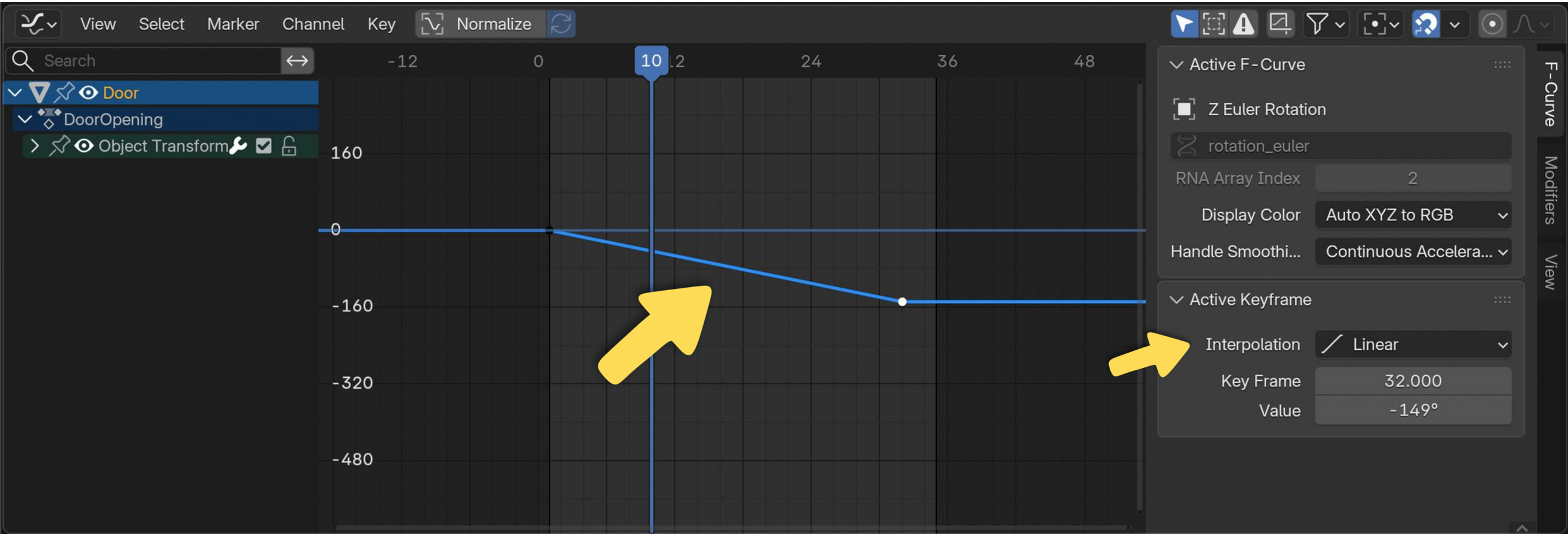
Interpolation



Interpolation | Constant



Interpolation | Linear

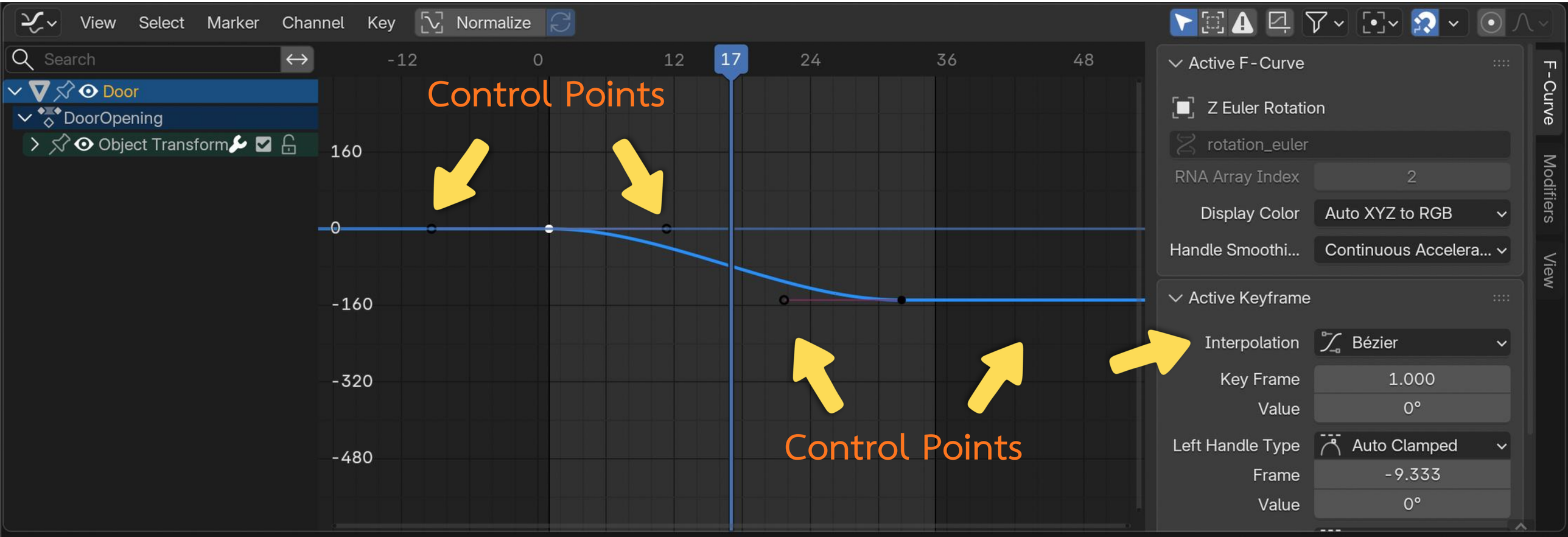


Interpolation | Linear

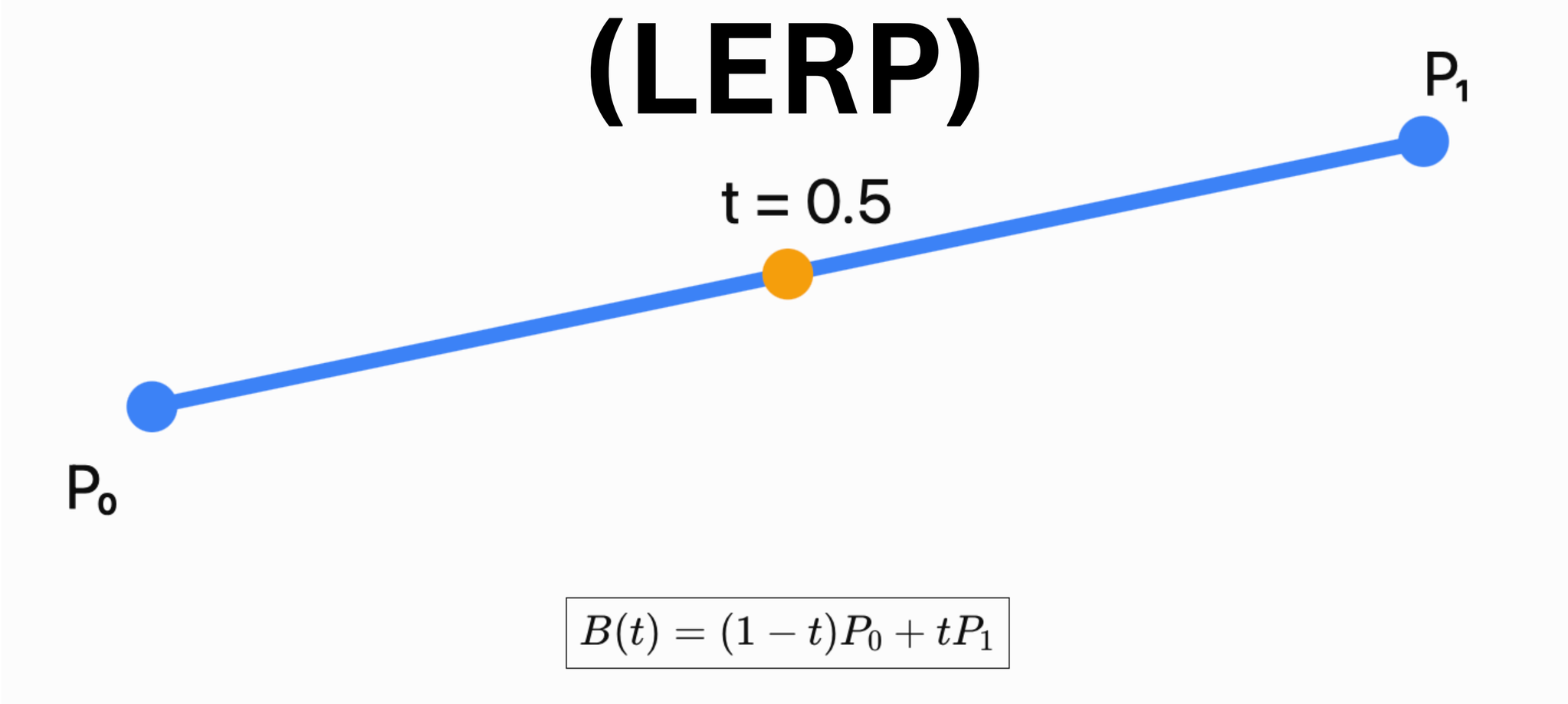




Interpolation | Bezier

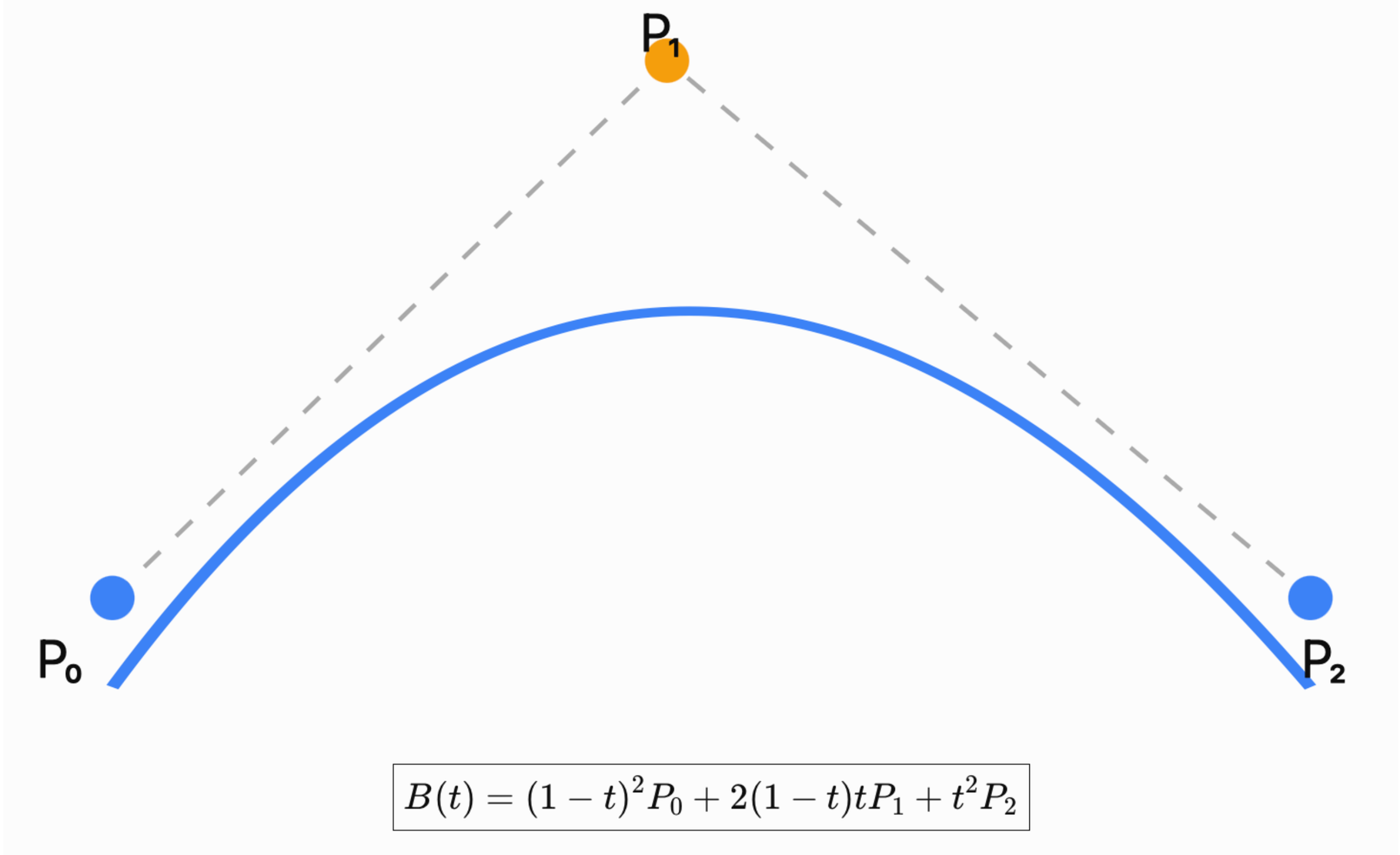


Linear Bezier 2 จุด (LERP)



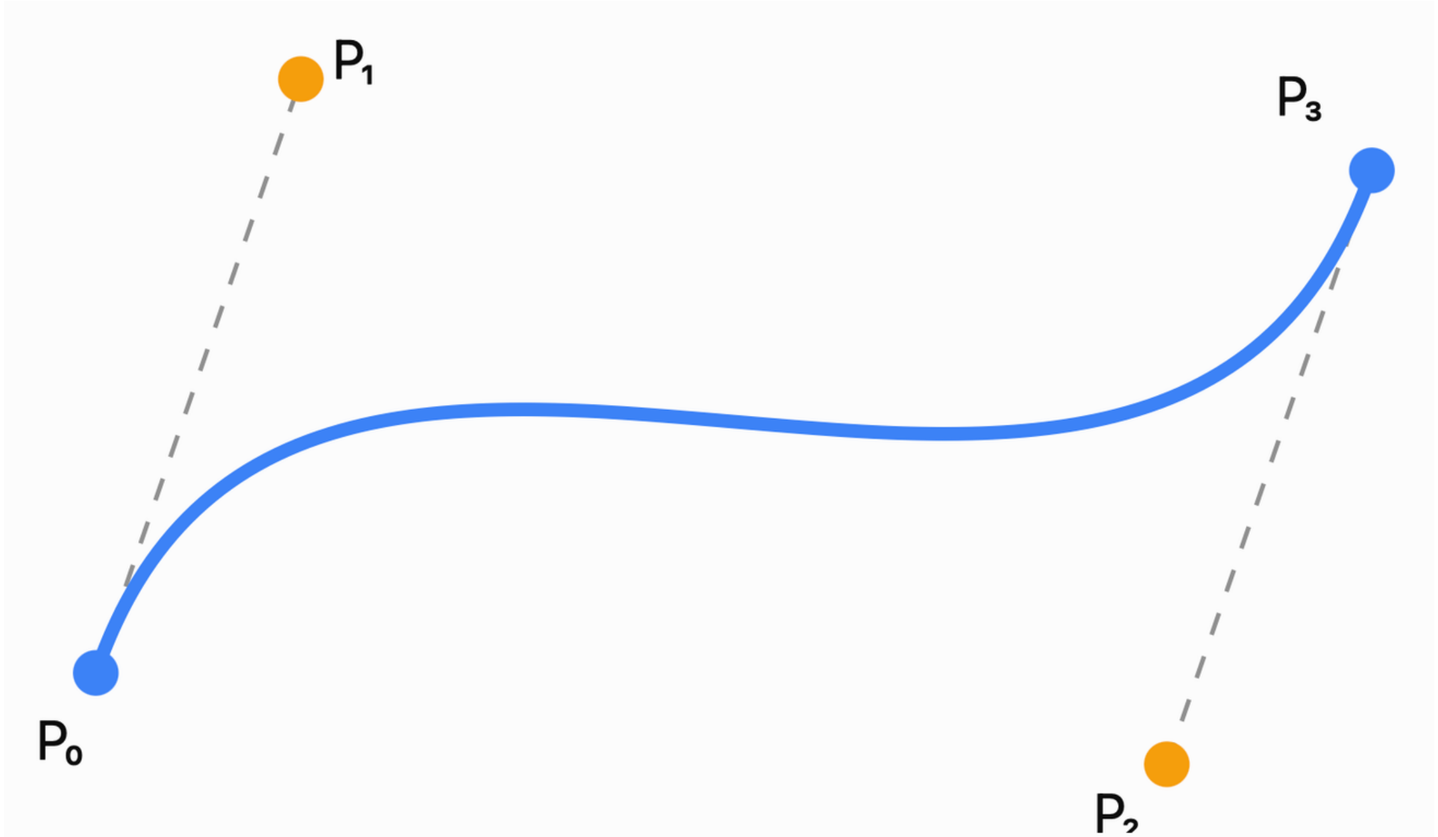
$$B(t) = (1-t)P_0 + tP_1$$

Quadratic Bézier (3 points)

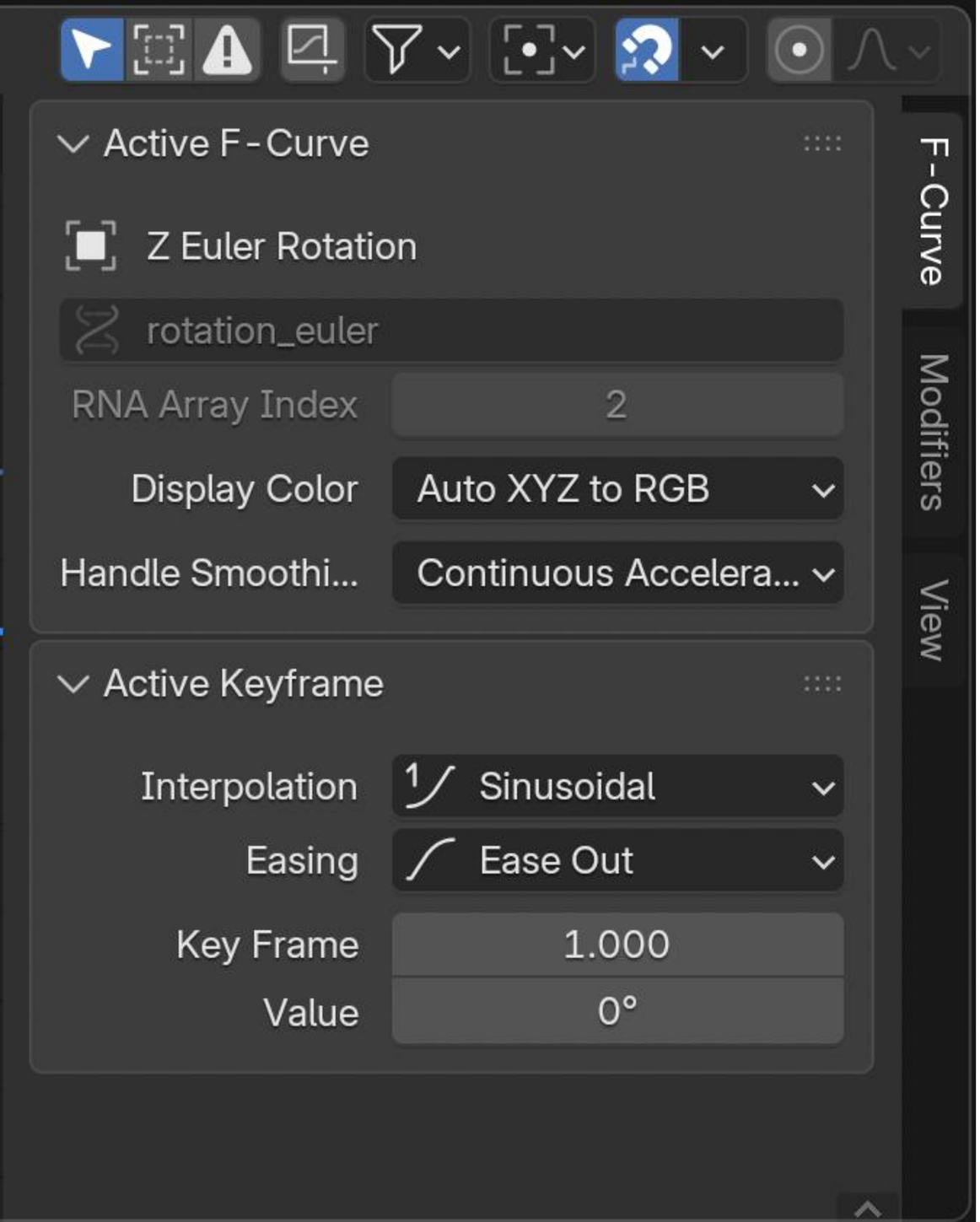


$$B(t) = (1-t)^2 P_0 + 2(1-t)tP_1 + t^2 P_2$$

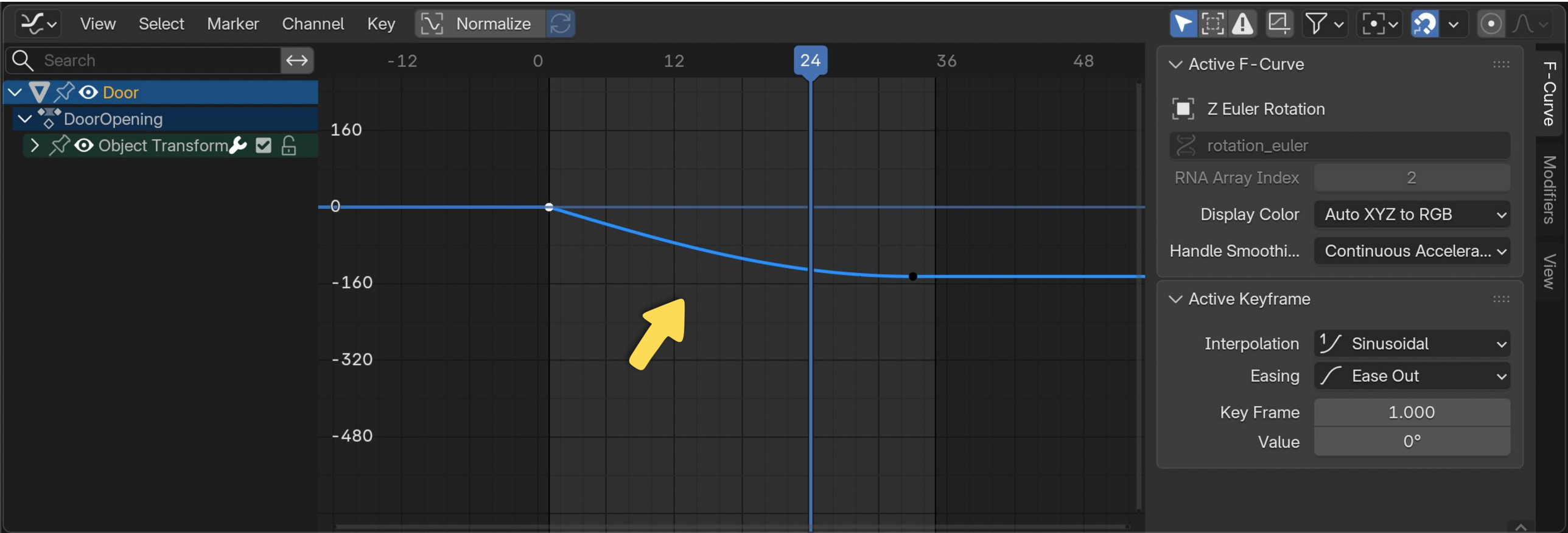
Cubic Bézier (4 points)



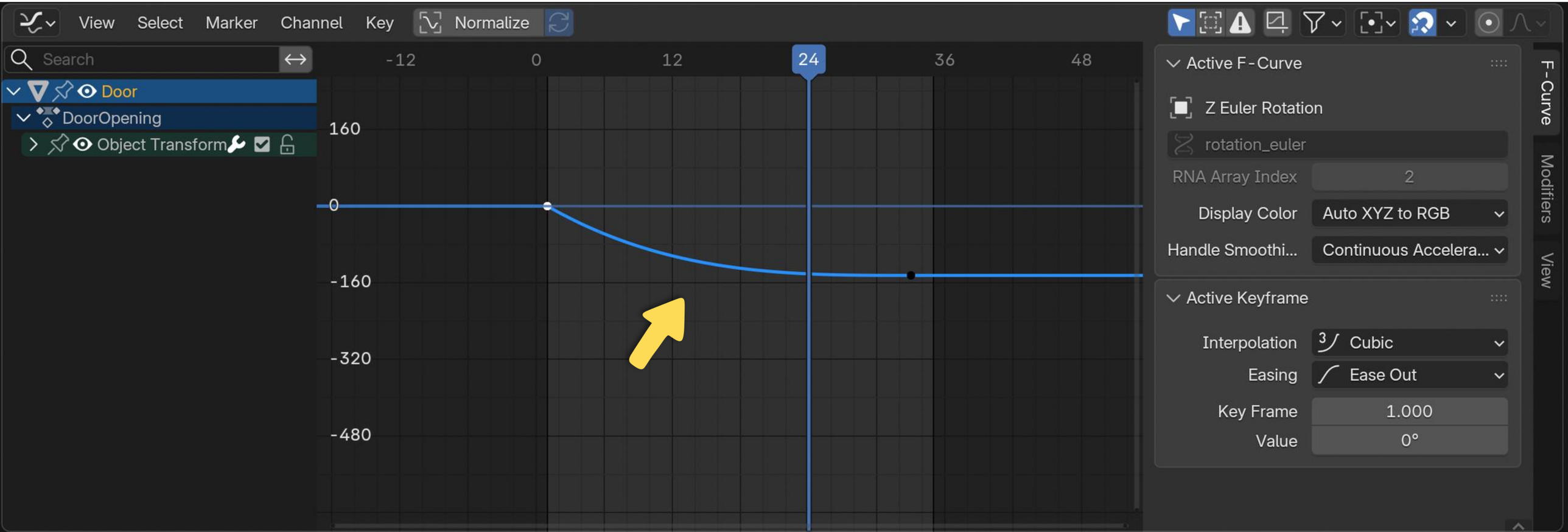
$$
\begin{aligned}
B(t) = {} & (1-t)^3 P_0 \\
& + 3(1-t)^2 t P_1 \\
& + 3(1-t) t^2 P_2 \\
& + t^3 P_3
\end{aligned}
$$


Active F-Curve
Z Euler Rotation
rotation_euler
RNA Array Index
2
Display Color
Auto XYZ to RGB
Handle Smoothi...
Continuous Accelera...
Active Keyframe
Interpolation
Sinusoidal
Easing
Ease Out
Key Frame
1.000
Value
0°
F-Curve
Modifiers
View

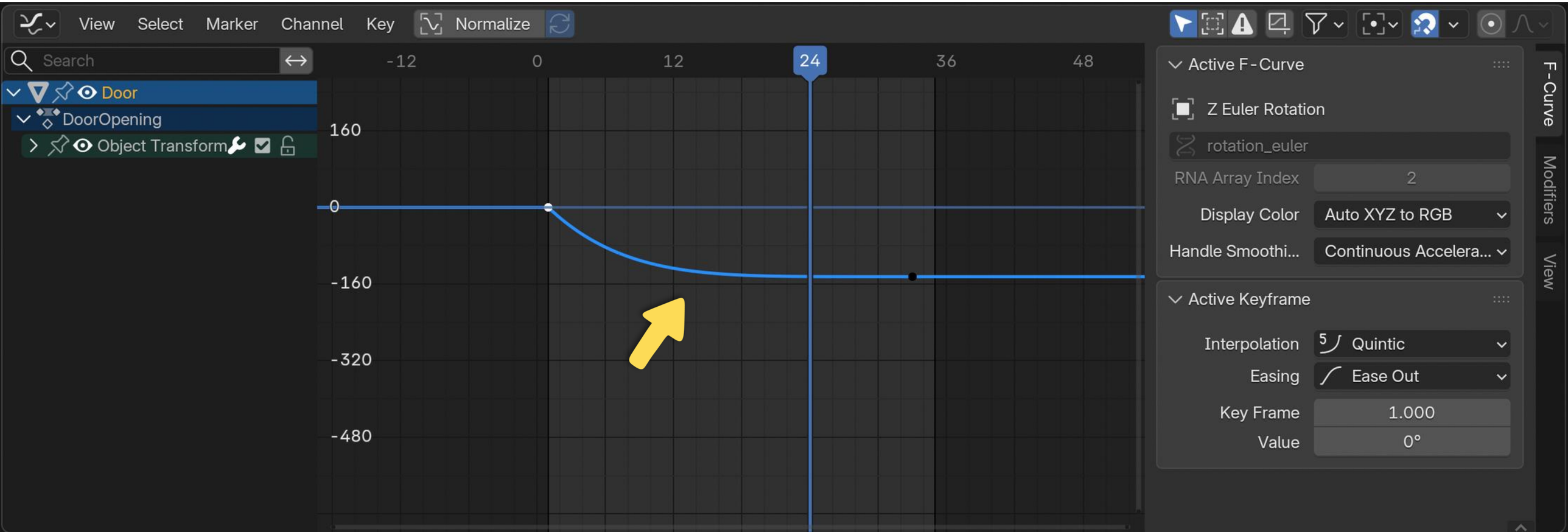
Interpolation | Sinusoidal



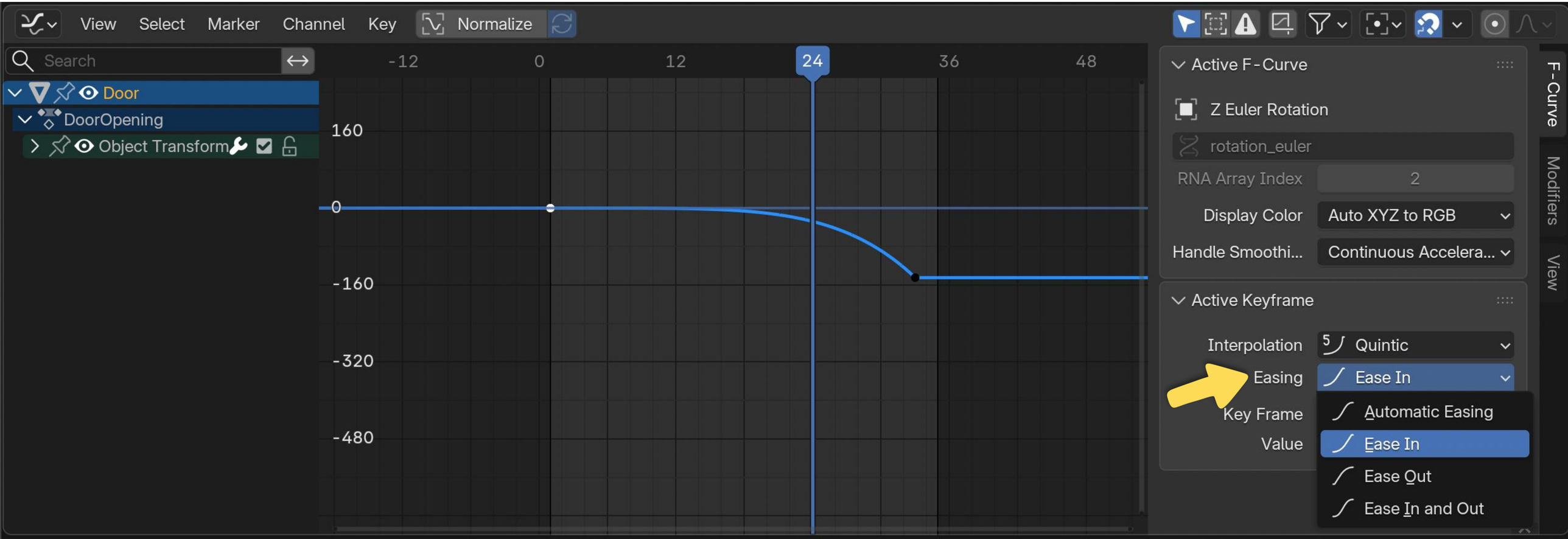
Interpolation | Cubic

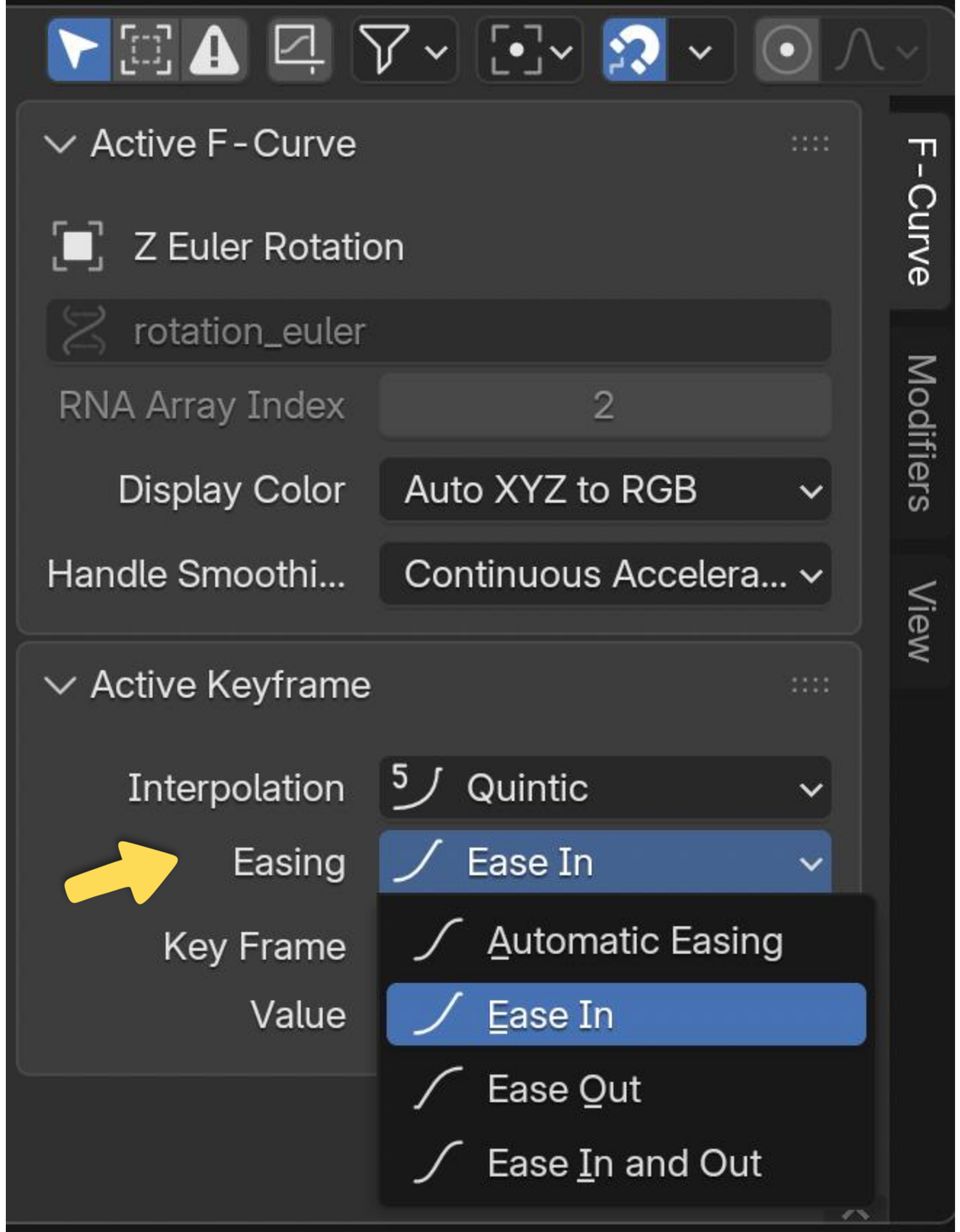


Interpolation | Quintic

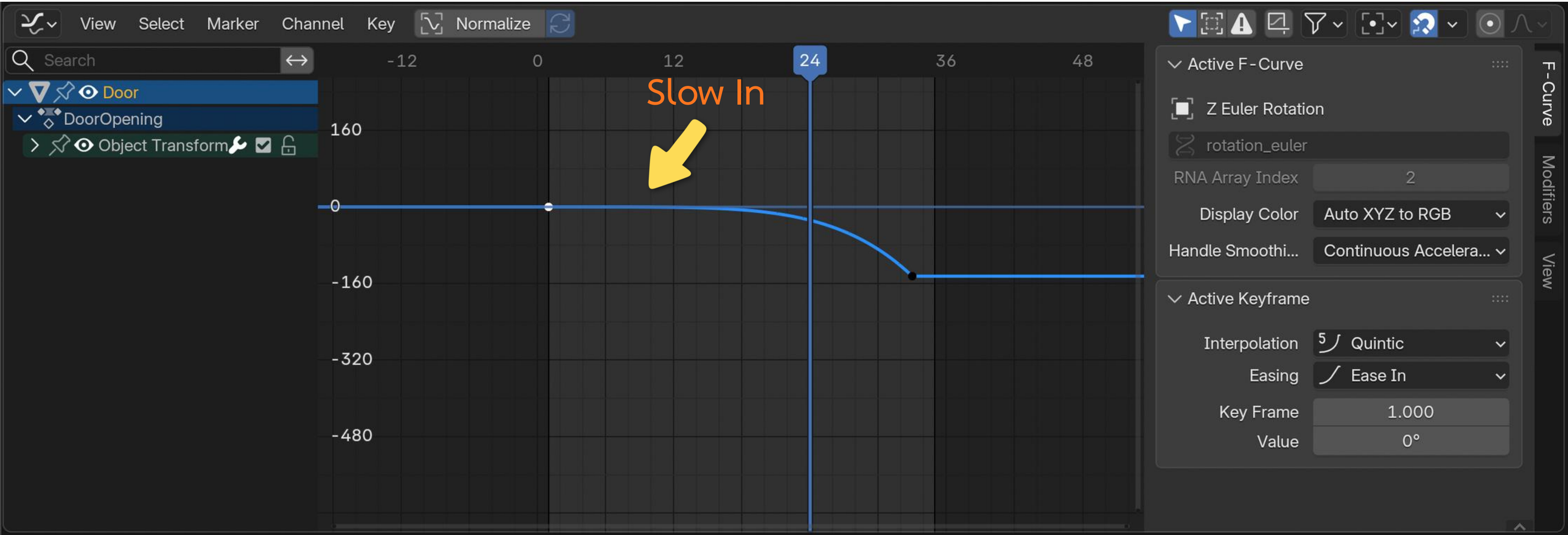


Easing

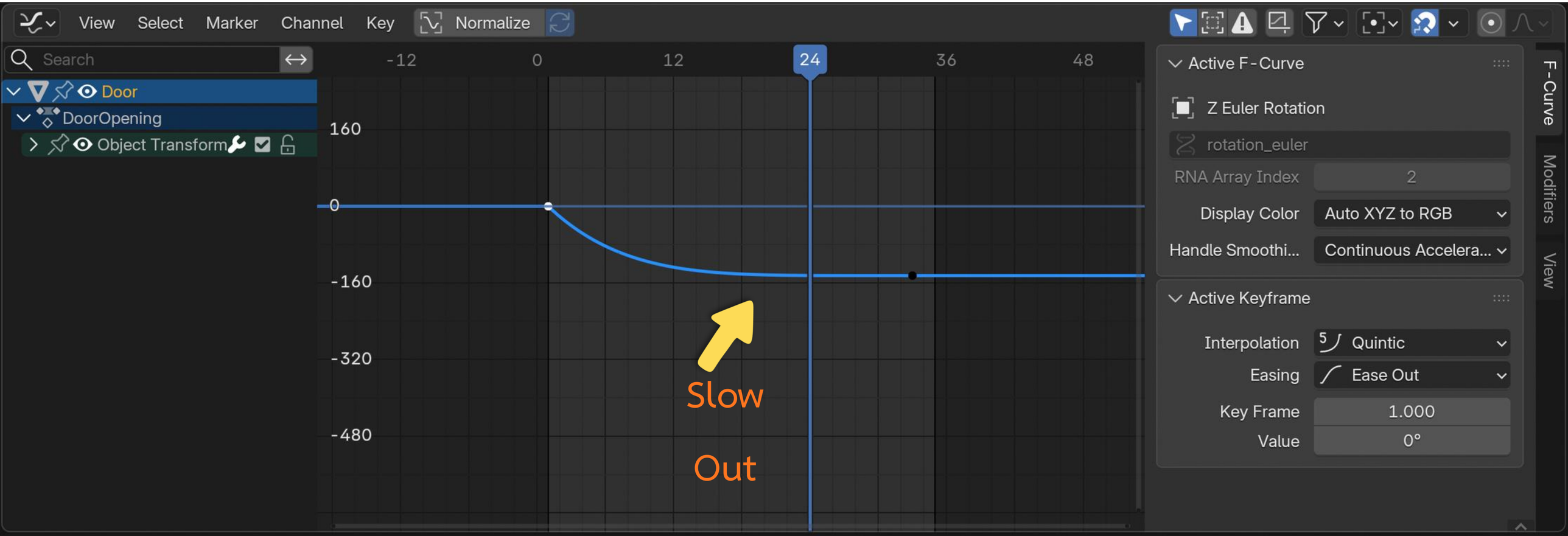

View
Select
Marker
Channel
Key
Normalize
Search
Door
DoorOpening
Object Transform
-12
0
12
24
36
48
160
0
-160
-320
-480
Active F-Curve
Z Euler Rotation
rotation_euler
RNA Array Index
2
Display Color
Auto XYZ to RGB
Handle Smoothi...
Continuous Accelera...
Active Keyframe
Interpolation
Quintic
Easing
Ease In
Key Frame
Value
Automatic Easing
Ease In
Ease Out
Ease In and Out
F-Curve
Modifiers
View


Active F-Curve
Z Euler Rotation
rotation_euler
RNA Array Index
2
Display Color
Auto XYZ to RGB
Handle Smoothi...
Continuous Accelera...
Active Keyframe
Interpolation
5
Quintic
Easing
Ease In
Key Frame
Value
Automatic Easing
Ease In
Ease Out
Ease In and Out
F-Curve
Modifiers
View

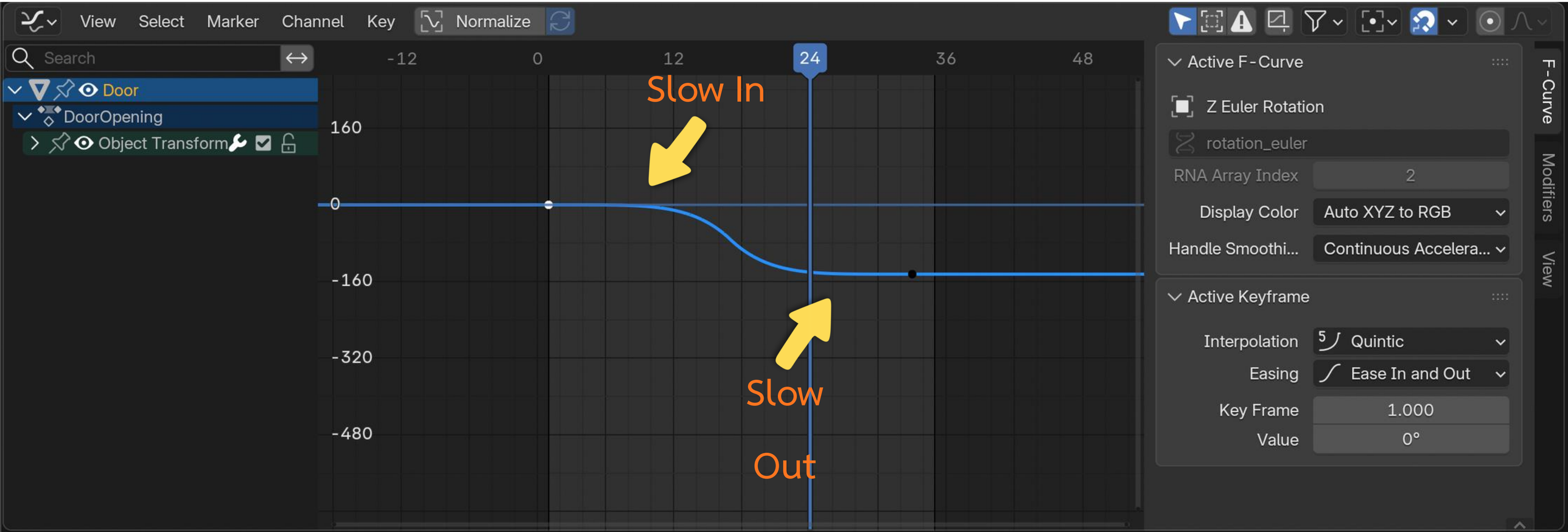
Easing | Ease In



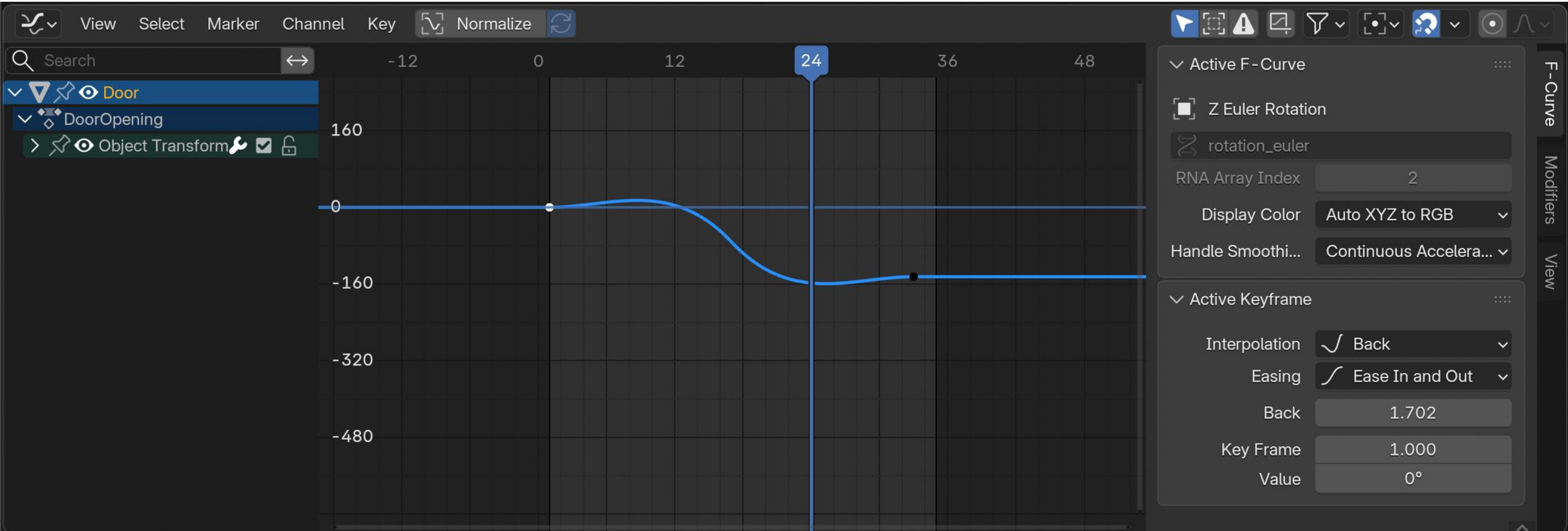
Easing | Ease Out



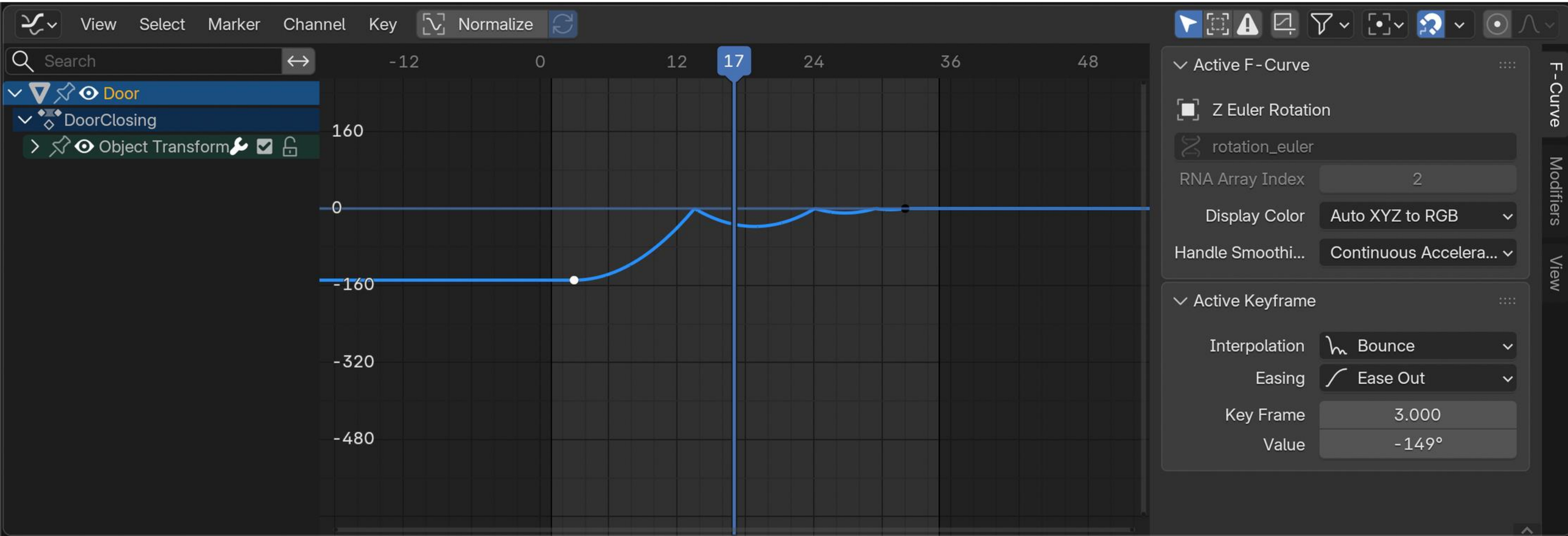
Easing | Ease InOut



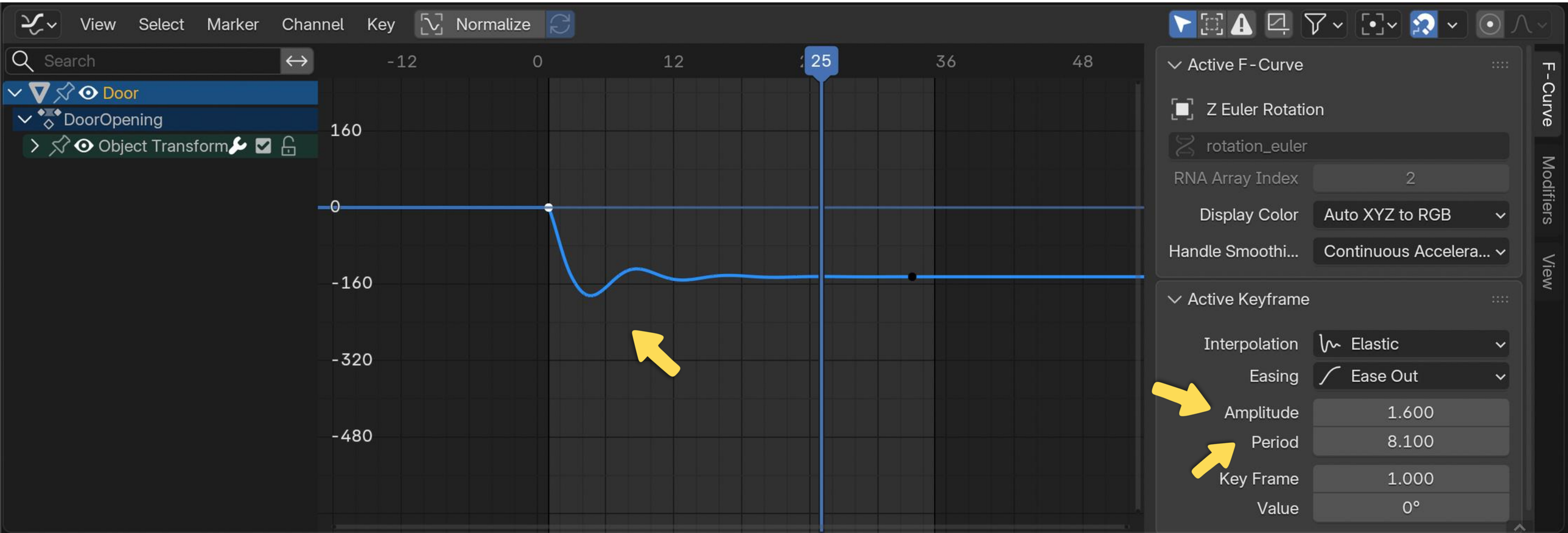
Effect | Back


View
Select
Marker
Channel
Key
Normalize
Search
Door
DoorOpening
Object Transform
-12
0
12
24
36
48
160
0
-160
-320
-480
Active F-Curve
Z Euler Rotation
rotation_euler
RNA Array Index
2
Display Color
Auto XYZ to RGB
Handle Smoothi...
Continuous Accelera...
Active Keyframe
Interpolation
Back
Easing
Ease In and Out
Back
1.702
Key Frame
1.000
Value
0°
F-Curve
Modifiers
View

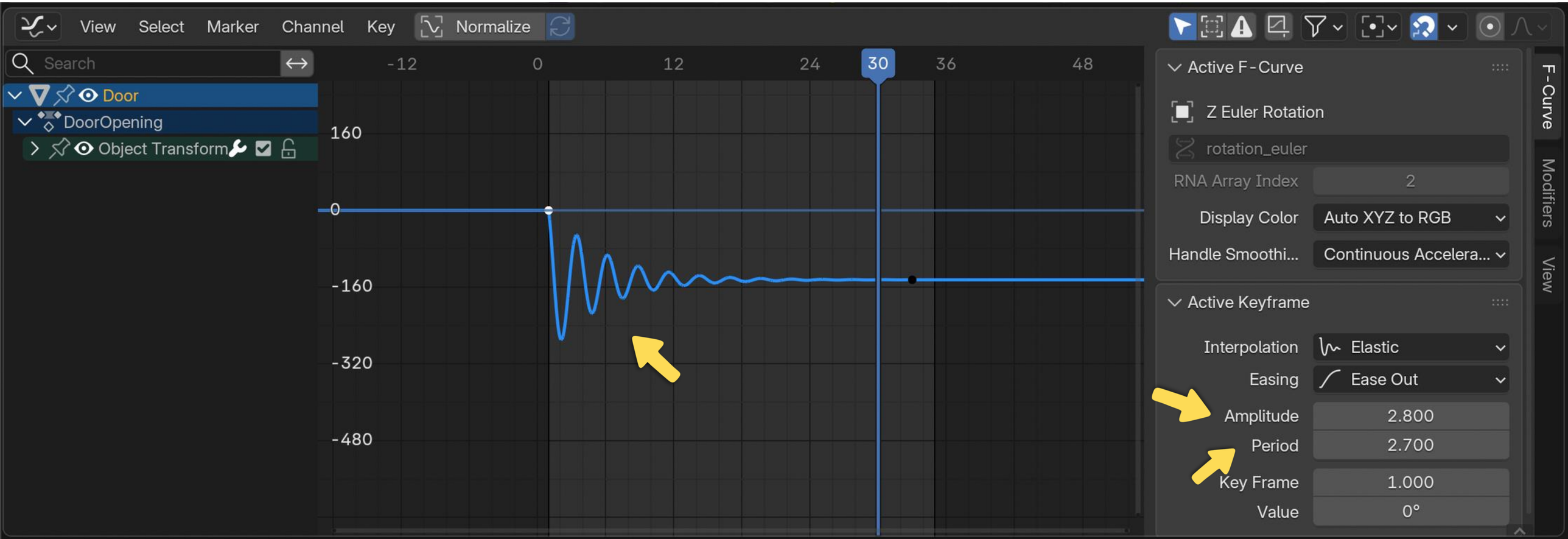
Effect | Bounce

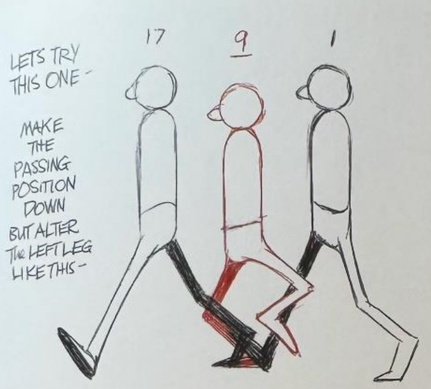


Effect | Elastic



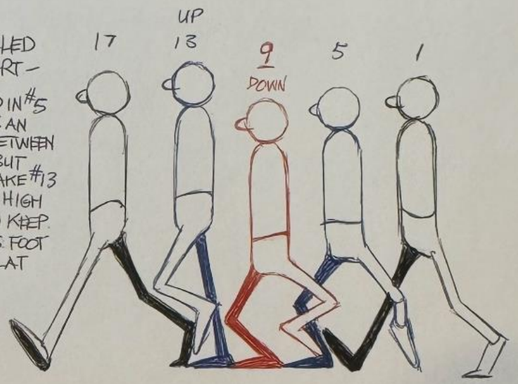
Effect | Elastic





LETS TRY THIS ONE -

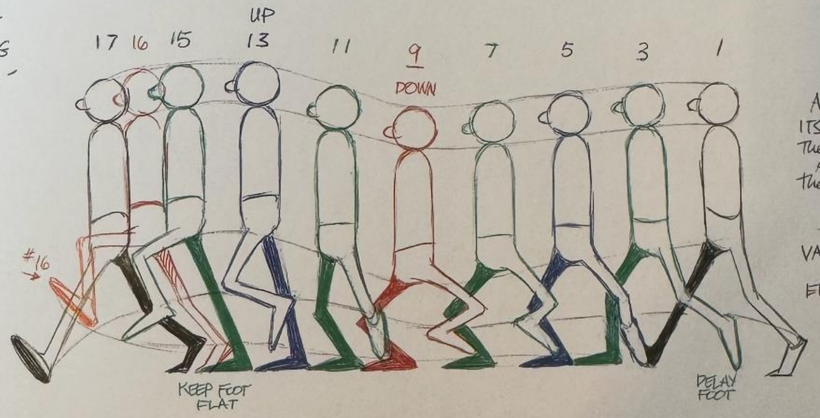
MAKE THE PASSING POSITION DOWN BUT ALTER THE LEFT LEG LIKE THIS -



PULLED APART -

ADD IN #5 AS AN IN BETWEEN BUT MAKE #13 THE HIGH 2nd KEEP IT'S FOOT FLAT

WE'D GET SOMETHING LIKE THIS -

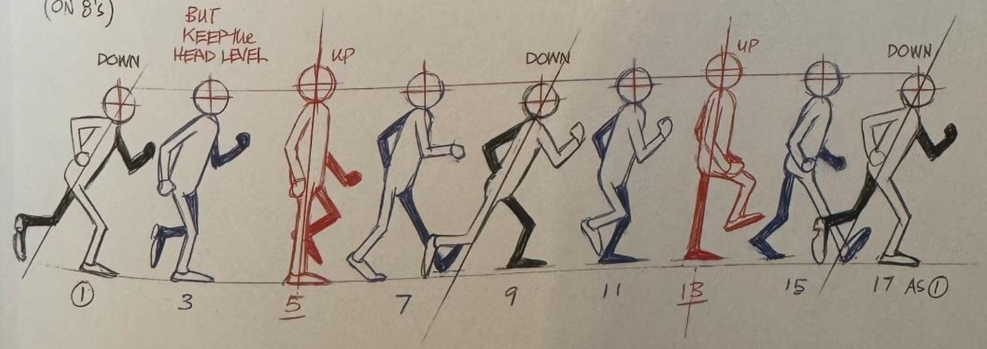


AGAIN, ITS ALL IN THE TIMING AND IN THE SPACING

AND THE VARIATIONS ARE ENDLESS.

LETS SIMPLY CHANGE the TILT OF THE BODY ON the PASSING POSITION -

(ON 8's)



LETS SIMPLY CHANGE the TILT OF the BODY ON the PASSING POSITION -

(ON 8's)

